
Lab 3

Advanced Data Manipulation

CSE 4508
RELATIONAL DATABASE MANAGEMENT SYSTEM LAB

OCTOBER 7, 2024

Contents

1	Join	2
2	String SQL Functions	4
3	Tasks	6

1 Join

Joins are used to combine rows from two or more tables based on related columns.

INNER JOIN

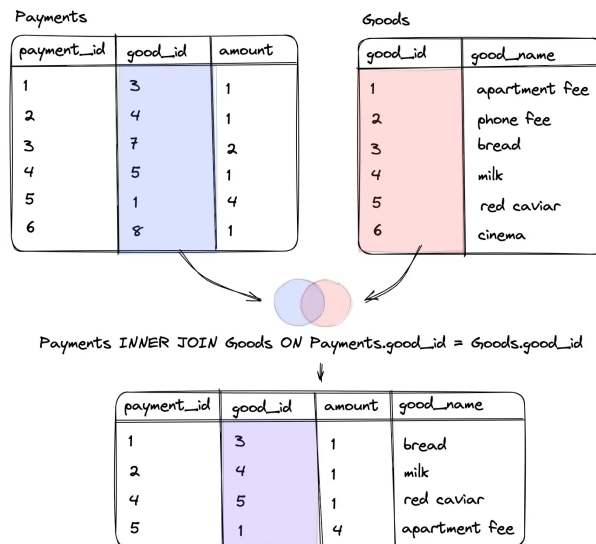


Figure 1: Sample of How Inner Join works

Another example:

```
SELECT S.STUDENT_NAME, C.COURSE_NAME
FROM STUDENT S
INNER JOIN ENROLLMENT E ON S.STUDENT_ID = E.STUDENT_ID
INNER JOIN COURSE C ON E.COURSE_ID = C.COURSE_ID;
```

This query aims to retrieve the names of students along with the courses they are enrolled in by joining three tables: **STUDENT**, **ENROLLMENT**, and **COURSE**.

S_ID	S_NAME	DEPT	S_ID	C_ID	C_ID	C_NAME	DEPT
1	Alice	CSE	1	101	101	Database Systems	CSE
2	Bob	EEE	2	102	102	Circuit Analysis	EEE
3	Carol	MCE	1	103	103	Operating Systems	CSE
			3	101			

Table 1: Student Table

Table 2: ENROLLMENT Table

Table 3: COURSE Table

Table 4: Before Tables

S_ID	S_NAME	DEPT	C_ID
1	Alice	CSE	101
1	Alice	CSE	103
2	Bob	EEE	102
3	Carol	MCE	101

Table 5: Intermediate Result (Table A)

S_ID	S_NAME	DEPT	C_ID	C_NAME
1	Alice	CSE	101	Database Systems
1	Alice	CSE	103	Operating Systems
2	Bob	EEE	102	Circuit Analysis
3	Carol	MCE	101	Database Systems

Table 6: Intermediate Result (Table B)

S_NAME	C_NAME
Alice	Database Systems
Alice	Operating Systems
Bob	Circuit Analysis
Carol	Database Systems

Table 7: Final Result: After INNER JOIN with COURSE

NATURAL JOIN

Natural join is an SQL join operation that creates a join on the base of the common columns in the tables. To perform natural join there must be one common attribute(Column) between two tables. Natural join will retrieve from multiple relations. It works in three steps.

```
SELECT *
FROM TABLE1
NATURAL JOIN TABLE2;
```

- A natural join matches columns with the same names in both tables and only includes rows where the values in those columns are equal (consistent tuples).
- It excludes rows where the values don't match (inconsistent tuples).
- After joining, it removes one of the duplicate columns from the result set.

```
SELECT *
FROM STUDENT
NATURAL JOIN ENROLLMENT;
```

S_ID	S_NAME	DEPT	S_ID	C_ID	C_ID	C_NAME	DEPT
1	Alice	CSE	1	101	101	Database Systems	CSE
2	Bob	EEE	2	102	102	Circuit Analysis	EEE
3	Carol	MCE	1	103	103	Operating Systems	CSE
			3	101			

Table 8: Student Table

Table 9: ENROLLMENT Table

Table 10: COURSE Table

Table 11: Before Tables

S_ID	S_NAME	DEPT	C_ID
1	Alice	CSE	101
1	Alice	CSE	103
2	Bob	EEE	102
3	Carol	MCE	101

Table 12: After Natural Join

2 String SQL Functions

1. **CONCAT**: Used to concatenate two or more strings.

```
SELECT CONCAT('Hello, ', 'World!') AS Greeting
FROM dual;
```

2. **INITCAP**: Capitalizes the first letter of each word in a string.

```
SELECT INITCAP('hello world') AS Proper_Case
FROM dual;
```

3. **LENGTH**: Returns the length of a string.

```
SELECT LENGTH('Hello, World!') AS String_Length
FROM dual;
```

4. **UPPER and LOWER**: Converts text to uppercase or lowercase.

```
SELECT UPPER('hello world') AS Uppercase_Text
FROM dual;
```

```
SELECT LOWER('HELLO WORLD') AS Lowercase_Text
FROM dual;
```

5. **SUBSTR**: Extracts a portion of a string.

```
SELECT SUBSTR('Hello, World!', 1, 5) AS Substring
FROM dual;
```

6. **INSTR**: Returns the position of a substring in a string.

```
SELECT INSTR('Hello, World!', 'World') AS Position
FROM dual;
```

7. **LPAD** and **RPAD**: Pads a string to the left or right with a specified character.

```
SELECT LPAD('Hello', 10, '*') AS Left_Padded
FROM dual;
```

```
SELECT RPAD('Hello', 10, '*') AS Right_Padded
FROM dual;
```

8. **LTRIM** and **RTRIM**: Removes leading or trailing spaces from a string.

```
SELECT LTRIM(' Hello') AS Trimmed_Left
FROM dual;
```

```
SELECT RTRIM('Hello ') AS Trimmed_Right
FROM dual;
```

9. **COUNT**: Counts the number of rows returned.

```
SELECT COUNT(*) AS Total_Rows
FROM dual;
```

3 Tasks

1. Run the .sql file to create the tables and insert data.
2. Display the Name, Customer ID, and Lifetime Usage of the top 5 highest users.
3. Display the name, date of birth, district, and division of all customers whose lifetime usage exceeds the average usage of Prepaid users with Silver status.
4. Count how many of the customers from the previous query are from Dhaka.
5. Write a query to display the full name of each customer, with the prefix "Mr./Ms." concatenated.
6. Write a query to display customer names with the first letter of each word capitalized.
7. Write a query to find customers whose names contain the substring "an". Return the position where this substring appears.
8. Write a query to display all customer names in lowercase.
9. Write a query to display all customer names in uppercase.
10. Write a query to display the length of each customer's name.
11. Write a query to display the customer names padded to the left with asterisks (*) to make the total length 15 characters.
12. Write a query to display the customer names after trimming any leading whitespace.
13. Write a query to display the first 5 characters of each customer's name.
14. Write a query to count the number of customers from each division.

Submission Guidelines

Submit a .sql file containing all your code.